

# Three Calendars, Three Kinds of Time

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## Abstract

In October 1582 ten calendar days were removed from the European year by decree, the better to bring the civil year back into step with the sun. That episode is used here as an entry point to the general claim that a calendar is a graduated scale imposed on the motion of the heavens. Three calendars are in ordinary use in China, and they are built on three different reference points: the tropical year, the synodic month, and the twenty-four solar terms. This note sets out the basis of each in turn, shows that the lunisolar calendar (农历, *nongli*) is neither a lunar nor a solar calendar but a compound of the two, and argues that the common habit of calling it the "lunar calendar" is not a loose usage but an error of classification. The stem-and-branch calendar (干支历, *ganzhili*) is treated as a solar calendar in its own right rather than as an appendage of the lunisolar one. A comparison table sets the four systems side by side. The note is descriptive; it contains no predictive claims.

**Keywords:** Chinese calendar; lunisolar calendar; tropical year; synodic month; solar terms; Gregorian reform; stem-and-branch calendar

## 1. A name that needs correcting

In October 1582, ten days disappeared from the European calendar. Thursday 4 October was followed directly by Friday 15 October. The intervening ten days were never lived through by anyone.

This was not a scribal error. It was an order issued by Pope Gregory XIII, and the reason was simple enough: the calendar had drifted out of step with the sun by ten days, so ten days were deleted.

The episode makes the nature of a calendar visible. A calendar is a graduated scale that human beings impose on the motion of the heavens. One revolution of the sun is taken as a year; one cycle of the moon's phases is taken as a month. The two periods do not divide evenly into one another, and no arrangement reconciles them completely. How the reconciliation is attempted is what separates one calendar from another.

Three calendars matter in ordinary Chinese life: the solar calendar, the lunar calendar, and the stem-and-branch calendar. Their reference points are entirely different, and mixing them produces errors.

Many people call the lunisolar calendar (农历) the "lunar calendar" (阴历). The name is inaccurate, and inaccurate by a wide margin.

Calendars fall into three classes by reference point.

- **Lunar.** The moon alone is observed. Months follow the phases; the sun's position is disregarded.
- **Solar.** The sun alone is observed. Years follow the tropical year; the phases are disregarded.
- **Lunisolar.** Both are observed. Months follow the moon, years follow the sun, and the difference between the two is made up by intercalary months.

The Chinese lunisolar calendar belongs to the third class. It fixes the first and fifteenth of each month by the new moon and full moon, and it keeps the year length in step with the tropical year by inserting intercalary months. Observing both, it is called a lunisolar calendar. The genuinely lunar calendar is the Islamic one, which follows the moon alone.

To call the lunisolar calendar "lunar" is to describe a hybrid as though it had a single parent.

## 2. The solar calendar: following the sun

The reference point of a solar calendar is the **tropical year**, the interval between two successive passages of the sun through the vernal equinox: 365.2422 days, or 365 days 5 hours 48 minutes 46 seconds.

The difficulty is that this number is not a whole number. Counting a year as 365 days loses 0.2422 of a day each year, and in four years the loss accumulates to nearly a full day. That day has to be put back, and the putting back is the leap year.

The method has taken wrong turnings. The Julian calendar, fixed by Julius Caesar in 45 BCE, simply inserted a leap day every four years, giving an average year of 365.25 days. That exceeds the tropical year by 0.0078 of a day, or 11 minutes 14 seconds a year. The figure looks negligible, but in 128 years it amounts to a full day.

By 1582 the accumulated error had reached ten days. The vernal equinox had moved from 21 March to 11 March. The Church cared about this, because the date of Easter is computed from the equinox, and a wandering equinox meant a wandering feast. The calendar was therefore reformed, in two moves.

First, **ten days were deleted**: the day after 4 October 1582 was declared to be 15 October, with the days of the week continuing unbroken.

Second, **the leap rule was changed**: a leap day every four years, none in century years, but one again in years divisible by 400. Thus 1700, 1800 and 1900 are not leap years, and 2000 is.

Under this rule only 97 of every 400 years are leap years, giving an average year of 365.2425 days, which differs from the tropical year by 0.0003 of a day, or 25.92 seconds a year. Accumulating a full day would take more than three thousand three hundred years.

The Gregorian calendar is the civil calendar now in use.

One distinction must be kept clear: **the lengths of the months in the civil calendar are arbitrary divisions and have nothing to do with the heavens**. The 365 days of the year are cut into twelve parts, and which month has 31 days and which has 30 is a historical convention. February is short, on one account, because Augustus moved a day from it to August. Neither the phases of the moon nor the position of the sun enters into the division.

This is the fundamental difference between the civil calendar and the stem-and-branch calendar discussed below.

### 3. The lunar calendar: following the moon

The reference point of a lunar calendar is the **synodic month**, the interval between two successive new moons: 29.5306 days, or 29 days 12 hours 44 minutes 3 seconds.

The moon does not shine by its own light. The phases we see are the changing illuminated half of the moon as viewed from the earth. From invisible, through full, and back to invisible takes twenty-nine and a half days. The period is visible to anyone who looks up, which is why the earliest calendars generally began from the phases.

The difficulty arises here as well: twelve synodic months amount to 354.3672 days, nearly eleven days short of the tropical year.

The purely lunar calendar's response is to leave the discrepancy alone.

The Islamic calendar works this way. Odd-numbered months have 30 days and even-numbered months 29, giving a year of 354 days. The remainder is absorbed by adding single days: eleven

of every thirty years are leap years, and a leap year adds a day at the end of the twelfth month, producing a year of 355 days. It adds leap days, and never intercalary months.

The consequence is that dates drift through the seasons. The Islamic year is about 10 days 21 hours shorter than the solar year, so a month is lost every three years, and after 32.6 solar years the Islamic calendar has completed one whole additional circuit. Ramadan therefore rotates through the seasons: in some years it falls in summer, when the days are long and the fast is hard; a decade or so later it falls in winter, when the days are short.

The advantage of the purely lunar calendar is that the phases and the dates always agree. The price is that dates and seasons come apart completely. For an agrarian society dependent on the heavens, that price is not payable.

#### 4. The stem-and-branch calendar: also solar, but divided differently

The stem-and-branch calendar is often treated as an appendage of the lunisolar calendar. It is in fact considerably more independent, and it is a purely solar calendar: it observes the sun and takes no account of the moon at all.

Its graduations are the twenty-four solar terms. The sun's path along the ecliptic is divided into steps of 15 degrees, and 360 degrees make twenty-four terms. The sequence opens with 立春 (Beginning of Spring) and closes with 大寒 (Great Cold). The ecliptic longitudes are fixed: 立春 stands at 315 degrees, each subsequent term adds 15 degrees, and the vernal equinox returns to 0 degrees.

Months are divided by the jie (节) and not by the qi (气). Among the twenty-four terms, the odd-numbered are jie and the even-numbered are qi; one jie paired with one qi makes a month, giving twelve months in the year, with no intercalary month required. A complete revolution of the sun is a year by definition and needs no correction.

The year of the month begins at 立春, and the branch is fixed:

- 立春 opens the 寅 month, 惊蛰 the 卯 month, 清明 the 辰 month
- 立夏 opens the 巳 month, 芒种 the 午 month, 小暑 the 未 month
- 立秋 opens the 申 month, 白露 the 酉 month, 寒露 the 戌 month
- 立冬 opens the 亥 month, 大雪 the 子 month, 小寒 the 丑 month

This is the familiar 正月建寅, "the first month is established at 寅". The 正月 here is the first month of the stem-and-branch calendar, beginning at 立春, which is not the same thing as the first day of the first month of the lunisolar calendar; that discrepancy was the subject of an earlier note in this series.

The key difference between the stem-and-branch calendar and the civil calendar lies here. Both are solar, and both follow the tropical year, but the months of the civil calendar are cut arbitrarily, whereas the months of the stem-and-branch calendar are cut by the sun's position. February may have 28 days or 29, depending on divisibility; the 寅 month is always the stretch from 立春 to 惊蛰, depending on where the sun has reached. One follows convention; the other follows the heavens.

The counting of days by stems and branches deserves one further remark. It is the longest continuously running day-count known: from the day 己巳 in the second month of the third year of Duke Yin of Lu in the Chunqiu (720 BCE) it has run for more than two thousand seven hundred years without a break. That starting point was recovered by back-calculation from an eclipse record: the Chunqiu entry 己巳, 日有食之, "on the day 己巳, the sun was eclipsed", corresponds by astronomical computation to 22 February 720 BCE.

A complete sexagenary table is already incised on Shang oracle bone: the piece excavated at Yinxu and catalogued as 合集 37986 contains all sixty designations without omission, and served as a reference table against which dates were checked.

## 5. Where the lunisolar calendar stands

Return to the question of the name. The lunisolar calendar is pure on neither side, and it combines the strengths of both:

The months follow the moon. The first of the month is always the day of the new moon, and the fifteenth is always near the full moon, so that the phase can be read off the date. This is the strength of the lunar calendar.

The years follow the sun. Intercalary months restore 354 days to 365, so the seasons do not wander. This is the strength of the solar calendar.

The twenty-four solar terms are borrowed from the stem-and-branch calendar to govern agricultural timing.

The traditional rule for making up the difference is seven intercalations in nineteen years, which is approximate. Which month receives the intercalation in a given year is determined by whether that month contains a 己. The procedure is now written into the national standard GB/T 33661-2017, computed and promulgated by the Purple Mountain Observatory.

The lunisolar calendar is therefore a compound of three things: the moon governs the month, the sun governs the year, and the solar terms govern agricultural timing. To call it the lunar calendar is to name one third of it.

## 6. Four calendars side by side

|                  | Solar (civil)         | Lunar (Islamic)                | Stem-and-branch             | Lunisolar                    |
|------------------|-----------------------|--------------------------------|-----------------------------|------------------------------|
| What it observes | Sun                   | Moon                           | Sun                         | Moon + Sun                   |
| Year length      | 365 / 366 days        | 354 / 355 days                 | c. 365.25 days              | 354 / 384 days               |
| Month length     | Arbitrary, 28–31 days | 29 / 30 days, by phase         | Two solar terms, c. 30 days | 29 / 30 days, by phase       |
| Correction       | Leap day              | Leap day, no intercalary month | None required               | Intercalary month            |
| Year begins      | 1 January             | 1 Muharram                     | 立春                          | First day of the first month |

The middle two columns are pure; the outer two are compromises.

## 7. What the distinction is for

The stem-and-branch calendar is used for setting out the four pillars: the day pillar and the month pillar of a birth time are determined entirely by the solar terms and have nothing to do with the first or fifteenth of a lunar month.

The civil calendar is used for daily life and for international correspondence; it counts the sun.

The lunisolar calendar is used for observing the phases and for festivals; a full moon on the fifteenth of the eighth month is something it governs.

Each of the three has its own province, and none can replace another. The consequence of mixing them is that a single date yields two results, and a dispute follows over which is right. Both are right; they are simply not the same calendar.

What comes next in this series is the solar terms themselves, where 立春 turns out not to be a whole day but a single second.

## Note on the text

This is a popular exposition of material treated in the author's teaching notes on the foundations of four-pillar practice, and it makes no claim to original research. The astronomical constants cited are standard values. This English version is issued alongside the Chinese original as the same article in another language. The author's name is 邝楚嘉, Chujia Kuang; he writes under the pen-name 嘉言一得 (Jiayan Yide).

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